26 The nth term of a sequence is $2n^2 - 1$

The nth term of a different sequence is $40 - n^2$

Show that there is only one number that is in both of these sequences.

$2n^2 - 1$: 1, 7, 17, 31, 49, 71, ...

$40 - n^2$: 39, 36, 31, 24, 15, 4, −9, ...

The first sequence increases; the second decreases.

They share 31 only (n = 4 and n = 3 respectively).

3 marks total
Calculate terms
At least 5 and 6 terms
Only 31 common

(Total for Question 26 is 3 marks)